Technical Supplement to Bayes-PC

Statistic-Level Bayes-Factor Mappings, Calibration, and Reproducibility

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Abstract

This separate technical note documents the adaptive conditional-independence (CI) evidence provider used to instantiate the evidence interface of Bayes-PC. It collects the test-selection protocol, the statistic-level null and alternative likelihoods, standardized-effect priors, concrete statistic-to-Bayes-factor mappings, the main numerical identities, and admissibility diagnostics. It expands the original working-paper appendix using the independently developed provider's technical derivations. It is an external reproducibility document, not an appendix to the IJTCS-FAW 2026 proceedings paper. All formulas below concern evidence models for CI statistics; Bayes-PC's collider-orientation rules are defined in the main paper.

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1 Scope, notation, and experimental interface

For a query $q = (X, Y, S)$, write $H_0(q) : X \perp Y \mid S$ and $H_1(q) : X \not\perp Y \mid S$. A separately trained and frozen selector recommends one CI procedure k using descriptors of the observed query. The selected CI test produces a statistic t_q; a test-specific evidence model subsequently returns

$$\ell(q) = \log \mathrm{BF}^{\mathrm{stat}}_{01,k}(q) = \log m_{0,k}(t_q; \mathcal{C}_{q,k}) - \log m_{1,k}(t_q; \mathcal{C}_{q,k}), \tag{1}$$

where $\mathcal{C}_{q,k}$ contains fixed or plug-in calibration values. Only the numerical evidence $\ell(q)$, not the selector's predicted reliability ranking, is consumed by Bayes-PC. Positive values favor CI relative to dependence and negative values favor dependence.

The discrete candidate library used in the conference experiments is Pearson's χ^2, G^2, PLLCIT, UCI, and wUCI; the continuous library consists of Fisher-Z, KCI, RCIT, and RCoT. Earlier provider research also studied LCIT and AdaRCIT, but those methods are not among the cited Bayes-PC experimental candidate configurations. They are therefore not required to explain the reported results.

1.1 Frozen adaptive selector

Separate selectors address discrete/continuous modalities and conditioning orders $|S| = 0$, $|S| = 1$, and $|S| \geq 2$. Features indicate potential test failure regimes: contingency sparsity, category imbalance, slice-level information and effective degrees of freedom for discrete data; nonlinear association, non-Gaussianity, heteroscedasticity, conditioning dimension and covariance geometry for continuous data. Offline training uses CI tasks with known ground truth, separate from the graph instances used for Bayes-PC evaluation.

For offline task i, every candidate test k is run and assigned a truth-aligned score

$$r_{ik} = s_i \ell_{ik}, \quad s_i = \begin{cases} +1, & H_0(q_i) \text{ true}, \\ -1, & H_1(q_i) \text{ true}, \end{cases} \qquad y_{ik} = \frac{r_{ik} - \min_j r_{ij}}{\max_j r_{ij} - \min_j r_{ij} + \varepsilon}, \tag{2}$$

with a small positive regularizer ε for coincident scores. A multi-output predictor maps observable task features to $\widehat{y}_{ik}$; the recommended procedure is $k^*(q) = \arg\max_k \widehat{y}_k(q)$. Only that candidate is executed online. The original provider configuration describes multi-output XGBoost selectors (learning rate 0.015, 5,000 estimators, depth 7, minimum child weight 5, row and feature subsampling each 0.8), trained on more than 10,000 independent CI tasks per order regime. These are historical frozen-provider specifications, not new Bayes-PC hyperparameters. The model artifacts and exact feature serialization must accompany any executable release.

2 Common statistical evidence construction

For selected test k, assume a null statistic density $f_{0,k}$ and a family of alternative densities $f_{1,k}(t \mid \delta)$ defined on the same observed statistic space. Let θ_k measure deviation from the CI null and $s_k > 0$ be a fixed method-specific reference scale. The dimensionless effect is $\delta_k = \theta_k / s_k$. Signed effects use $\delta \sim \mathrm{Cauchy}(0, r)$, whereas nonnegative effects use $\delta \sim \text{Half-Cauchy}(0, r)$, with densities

$$\pi_{\pm}(\delta; r) = \frac{1}{\pi r[1 + (\delta/r)^2]}, \quad -\infty < \delta < \infty; \qquad \pi_{+}(\delta; r) = \frac{2}{\pi r[1 + (\delta/r)^2]}, \quad \delta > 0. \tag{3}$$

The main implementation uses $r = 1$ after standardization. The marginal alternative likelihood and statistic-level Bayes factor are

$$m_{1,k}(t; \mathcal{C}_k) = \int_{\Omega_k} f_{1,k}(t \mid \delta; \mathcal{C}_k)\pi_k(\delta)\, d\delta, \qquad \mathrm{BF}^{\mathrm{stat}}_{01,k}(t) = \frac{f_{0,k}(t; \mathcal{C}_k)}{m_{1,k}(t; \mathcal{C}_k)}. \tag{4}$$

This construction provides a common evidence *direction* and prior convention, while the models $f_{0,k}, f_{1,k}$ remain test-specific. If $\mathcal{C}_k$ is estimated from the query data, the expression is a plug-in ratio of statistic-level predictive models. A shared BF scale does not establish equal frequentist errors under all models or conditioning orders.

2.1 Unit effect scale for root-N estimators

Suppose an estimator $\widehat{\theta}_{k,N}$ satisfies $\sqrt{N}(\widehat{\theta}_{k,N} - \theta_k) \Rightarrow \mathcal{N}(0, V_k)$, where $0 < V_k < \infty$, and the squared centered statistics are uniformly integrable. Then

$$N \operatorname{Var}(\widehat{\theta}_{k,N}) \longrightarrow V_k, \qquad s_k = \sqrt{V_k} = \lim_{N\to\infty} \sqrt{N}\, \operatorname{sd}(\widehat{\theta}_{k,N}). \tag{5}$$

Indeed, convergence in distribution plus uniform integrability gives convergence of first and second moments of the centered $\sqrt{N}$ statistic, and the variance identity follows. A ratio-consistent standard-error estimator yields the plug-in scale $\sqrt{N}\,\widehat{\mathrm{SE}}$ by Slutsky's theorem. This argument motivates standardized effects for regular estimators; it does not assert that every kernel quadratic form obeys the same ordinary root-N model. The latter receive their own moment-based scales in Sections 4–5.

2.2 Summary of test-specific models

Table 1: Evidence models in the Bayes-PC experimental CI provider. The alternative distributions for quadratic statistics are surrogate models preserving nonnegative support.

CI test	Statistic-level null	Conditional alternative and effect
Fisher-Z	Fisher-transform Gaussian	Gaussian mean shift; signed Cauchy prior; Voigt marginal
KCI	Gamma moment fit (scaled χ^2)	Noncentral χ^2, mean-offset effect; half-Cauchy
RCIT/RCoT	Weighted quadratic-form Gamma fit	Noncentral χ^2, moment-matched mean shift; half-Cauchy
Pearson χ^2/G^2	Empirically fitted Gamma	Noncentral $\chi^2_d(N\omega^2)$; half-Cauchy
PLLCIT	Central χ^2_d	Noncentral $\chi^2_d(N\omega^2)$; half-Cauchy
UCI/wUCI	Conditional permutation, skew-normal fit	Gaussian mean shift; bootstrap scale; half-Cauchy

3 Fisher-Z: full Gaussian–Cauchy marginalization

Let n be the sample size and S the conditioning set, and assume $n_e = n - |S| - 3 > 0$. With sample partial correlation $\widehat{\rho}_{XY\cdot S}$ and population partial correlation ρ, define

$$z = \operatorname{arctanh}(\widehat{\rho}_{XY\cdot S}), \quad \zeta = \operatorname{arctanh}(\rho), \quad T = \sqrt{n_e}\, z, \quad \delta = \zeta. \tag{6}$$

The Fisher-transform Gaussian approximation is $z \mid \zeta \approx \mathcal{N}(\zeta, 1/n_e)$; hence $T \mid \delta, H_1 \sim \mathcal{N}(\sqrt{n_e}\delta, 1)$ and $T \mid H_0 \sim \mathcal{N}(0, 1)$. Thus $m_0(T) = \phi(T)$. With $\delta \sim \mathrm{Cauchy}(0, r)$, the alternative is

$$m_1(T) = \int_{-\infty}^{\infty} \phi(T - \sqrt{n_e}\delta) \frac{d\delta}{\pi r[1 + (\delta/r)^2]}. \tag{7}$$

Set $u = \sqrt{n_e}\delta$. Cauchy scale equivariance gives $u \sim \text{Cauchy}(0, r\sqrt{n_e})$, so this integral is the Gaussian–Cauchy convolution

$$m_1(T) = \int_{-\infty}^{\infty} \phi(T-u)\frac{r\sqrt{n_e}}{\pi(u^2 + r^2 n_e)}\,du = V(T; 1, r\sqrt{n_e}), \tag{8}$$

where $V(t; \sigma, \gamma)$ is a Voigt density with Gaussian standard deviation σ and Cauchy scale γ. Therefore

$$\boxed{\ell_Z(q) = \log\phi(T) - \log V(T; 1, r\sqrt{n_e}).} \tag{9}$$

For implementation, the Voigt density has the special-function form

$$V(t; \sigma, \gamma) = \frac{\Re\left[w\left(\frac{t+i\gamma}{\sigma\sqrt{2}}\right)\right]}{\sigma\sqrt{2\pi}},$$

where w is the Faddeeva function. This is an exact convolution *conditional on the Fisher Gaussian sampling approximation.* Numerically, protect the calculation against $|\widehat{\rho}| = 1$ and $n_e \le 0$ by following the fixed implementation's admissibility and regularization rules; the uploaded technical description does not specify a universal clipping constant.

4 KCI: Gamma moments and noncentral-χ^2 evidence

Let $S_{\text{obs}} \ge 0$ be a KCI quadratic statistic. The weighted-χ^2 null is approximated by $S \mid H_0 \approx \text{Gamma}(k_0, \theta_0)$, with shape $k_0 > 0$ and scale $\theta_0 > 0$ obtained from its first two null moments. Transform $Y = 2S/\theta_0$, so $Y \mid H_0 \approx \chi^2_{2k_0}$.

Set $\Delta = \mathbb{E}_{H_1} S - \mathbb{E}_{H_0} S > 0$, $s_{\text{KCI}} = \sqrt{n}\,\theta_0\sqrt{k_0}$, and $\delta = \Delta/s_{\text{KCI}}$. The surrogate alternative $Y \mid \delta, H_1 \sim \chi^2_{2k_0}(\lambda(\delta))$ is fixed by matching the mean shift:

$$\frac{\theta_0}{2}\lambda(\delta) = \Delta, \qquad \lambda(\delta) = 2\delta\sqrt{nk_0}. \tag{10}$$

Writing $f_{\chi^2}(y; \nu, \lambda)$ for the noncentral chi-square density, the mapping is

$$m_0(y) = f_{\chi^2}(y; 2k_0, 0), \qquad m_1(y) = \int_0^{\infty} f_{\chi^2}(y; 2k_0, 2\delta\sqrt{nk_0})\frac{2\,d\delta}{\pi(1+\delta^2)}. \tag{11}$$

The same nonnegative statistic support is retained, and the conditional alternative nests the central model at $\delta = 0$.

5 RCIT and RCoT: quadratic-form moment matching

For the randomized conditional-independence statistics, let $S_{\text{obs}} = n\|\widehat{C}\|_F^2$ be the residual cross-covariance quadratic form. Under independence, the reference approximation is $S \overset{d}{\approx} \sum_j \lambda_j Z_j^2$ with $Z_j \sim \mathcal{N}(0,1)$ independent. Define

$$c_1 = \sum_j \lambda_j, \quad c_2 = 2\sum_j \lambda_j^2, \quad a_0 = \frac{c_1^2}{c_2}, \quad b_0 = \frac{c_2}{c_1}. \tag{12}$$

Moment matching gives $S \mid H_0 \approx \text{Gamma}(a_0, b_0)$. For $Y = 2S/b_0$, this implies $Y \mid H_0 \approx \chi^2_{2a_0}$. Define the alternative mean displacement $\Delta = \mathbb{E}_{H_1} S - c_1 > 0$, the reference scale $s_{\text{RCIT}} = \sqrt{nc_2}$, and the standardized effect $\delta = \Delta/s_{\text{RCIT}}$. Matching $\Delta = (b_0/2)\lambda$ for the noncentral χ^2 surrogate yields

$$\lambda(\delta) = 2\delta\sqrt{na_0}, \quad m_1(y) = \int_0^{\infty} f_{\chi^2}(y; 2a_0, 2\delta\sqrt{na_0})\frac{2\,d\delta}{\pi(1+\delta^2)}. \tag{13}$$

Here $m_0(y) = f_{\chi^2}(y; 2a_0, 0)$, and RCIT versus RCoT changes the feature representation and induced moments, not this statistic-level evidence formula.

6 Discrete divergence models: Pearson, G^2, and PLLCIT

For a nonnegative observed statistic s with effective degrees of freedom d, the documented null models are

$$\begin{aligned} m_0^{\text{Pearson}}(s) &= f_\Gamma(s; k_0, \theta_0), \\ m_0^{G^2}(s) &= f_\Gamma(s; k_0, \theta_0), \\ m_0^{\text{PLLCIT}}(s) &= f_{\chi^2}(s; d, 0). \end{aligned} \tag{14}$$

The Gamma null is fitted from an empirical reference statistic distribution in the discrete CI tasks; PLLCIT retains a central χ^2_d null. Define a dimensionless nonnegative divergence effect $\omega \geq 0$ and set the noncentrality $\lambda(\omega) = n\omega^2$. The common alternative mixture is

$$m_1(s) = \int_0^\infty f_{\chi^2}(s; d, n\omega^2) \frac{2\, d\omega}{\pi(1+\omega^2)}, \qquad \ell(s) = \log m_0(s) - \log m_1(s). \tag{15}$$

For PLLCIT the $\omega = 0$ model nests its central χ^2 null. For Pearson/G^2, the fitted Gamma null need not coincide with the $\omega = 0$ central-χ^2 model, so the ratio compares two specified (possibly nonnested) statistic-level models.

7 UCI and wUCI: permutation null and bootstrap alternative

Let T be the UCI/wUCI fourth-order U statistic. The null may be degenerate, so an empirical distribution is obtained by permutations *within conditioning strata* and approximated by a skew-normal model. With skewness parameter α, location ξ, and scale $\omega_{\text{sn}} > 0$, its density is

$$m_0(t) = \frac{2}{\omega_{\text{sn}}} \phi\left(\frac{t-\xi}{\omega_{\text{sn}}}\right) \Phi\left(\alpha \frac{t-\xi}{\omega_{\text{sn}}}\right). \tag{16}$$

Under a nondegenerate alternative, a root-n Gaussian statistic approximation is used. Let $\Delta > 0$ be the effect, and let $\sigma_t > 0$ be its bootstrap standard error at the given n. Define $\delta = \Delta/(\sqrt{n}\sigma_t)$ and model $T \mid \delta, H_1 \sim \mathcal{N}(\delta\sqrt{n}\sigma_t, \sigma_t^2)$. With half-Cauchy effect prior,

$$m_1(t) = \int_0^\infty \frac{\exp[-(t - \delta\sqrt{n}\sigma_t)^2/(2\sigma_t^2)]}{\sqrt{2\pi}\sigma_t} \frac{2\, d\delta}{\pi(1+\delta^2)}. \tag{17}$$

Equivalently one may integrate over the dimensional effect $\Delta = \delta\sqrt{n}\sigma_t$ using the transformed prior $\pi_\Delta(\Delta) = 2s/[\pi(\Delta^2 + s^2)]$, where $s = \sqrt{n}\sigma_t$. Both parameterizations yield the same marginal likelihood by change of variables. As with Pearson/G^2, the skew-normal empirical null and Gaussian alternative at zero effect are generally different model families.

8 Normalization, numerical stability, and interpretation

Proposition 1 (Proper statistic-level evidence ratio)**.** *Suppose f_0 is a proper null density, $f_1(t \mid \delta)$ is a proper conditional density for each δ, and $\pi(\delta)$ is a proper prior. Then $m_1(t) = \int f_1(t \mid \delta)\pi(\delta)d\delta$ is a normalized predictive density. Wherever $m_0(t)$ and $m_1(t)$ are positive and finite with respect to a shared dominating measure, $m_0(t)/m_1(t)$ is a well-defined statistic-level Bayes factor.*

Proof. By Tonelli's theorem, $\int m_1(t)dt = \int \pi(\delta)[\int f_1(t \mid \delta)dt]d\delta = 1$. The density ratio is therefore defined at every point satisfying the stated positivity and finiteness conditions. □

The statement concerns *internal normalization of the specified models*; it does not prove that an approximate statistic distribution matches the unknown finite-sample data-generating law. For a plug-in fit of null parameters or covariance moments, the likelihood ratio is conditional on the fitted configuration.

8.1 What a common evidence scale does and does not imply

The common $\log \mathrm{BF}_{01}$ convention supplies a dimensionless model-evidence *direction*, not an assertion that the CI tests have equal operating characteristics. One mathematical sufficient condition for a stronger interpretation is that all statistic-level null and alternative models are induced by the same two predictive probability measures P_0, P_1 on the original dataset D; this condition is not established merely by fixing a common prior scale.

Proposition 2 (Common predictive-model criterion)**.** *Suppose $P_0 \sim P_1$ are common prior-predictive models on the data D, $T_k = g_k(D)$ is a measurable statistic, and the statistic models $Q_{h,k} = P_h \circ g_k^{-1}$ admit the density ratio $L_k(T_k) = dQ_{0,k}/dQ_{1,k}(T_k)$. For $L(D) = dP_0/dP_1(D)$,*

$$L_k(T_k) = \mathbb{E}_{P_1}[L(D) \mid T_k] \quad (P_1\text{-a.s.}). \tag{18}$$

Proof. For any measurable subset A of the statistic space, $\mathbb{E}_{P_1}[L(D)\mathbf{1}_{\{T_k \in A\}}] = P_0(T_k \in A) = \mathbb{E}_{P_1}[L_k(T_k)\mathbf{1}_{\{T_k \in A\}}]$. The conditional-expectation identity follows. □

In the actual provider, fitted Gamma/skew-normal nulls, noncentral χ^2 surrogate alternatives, and plug-in numerical parameters need not be push-forwards of one pair of common P_0, P_1 models. Thus the proposition does *not* establish comparable finite-sample calibration for those implementations. It identifies the additional modeling assumption needed for a common-predictive interpretation, rather than assuming it.

8.2 Ideal-model bounds and adaptive post-selection

For *fixed* properly specified predictive densities m_0, m_1 with $m_0 \ll m_1$ and $m_1 \ll m_0$, idealized statistic-level Bayes factors obey

$$\Pr_{m_1}(\log \mathrm{BF}_{01} > \tau) \le e^{-\tau}, \qquad \Pr_{m_0}(\log \mathrm{BF}_{01} < -\tau) \le e^{-\tau}. \tag{19}$$

Indeed, $\mathbb{E}_{m_1}(\mathrm{BF}_{01}) = 1$ and $\mathbb{E}_{m_0}(\mathrm{BF}_{10}) = 1$, so both claims follow from Markov's inequality. These are bounds under the *specified prior-predictive distributions*, not frequentist error guarantees uniform over individual dependence effects or heterogeneous CI tasks. They also need not apply to plug-in approximations.

Crucially, the adaptive selector chooses $K = K(D)$ using descriptors computed from the same query data. Conditioning on $K = k$ generally changes the statistic law:

$$f^{\mathrm{sel}}_{h,k}(t) = p(T_k = t \mid K(D) = k, H_h) \ne f_{h,k}(t), \quad h \in \{0, 1\}, \tag{20}$$

where the right-hand side refers to the fixed-test statistic model. The implemented $\ell(q)$ does not include a selective-likelihood correction. Therefore the validity of Eq. (19) *after* test selection must not be presumed; an empirical post-selection audit is necessary. Conditioning order, empirical null fits, random-feature seeds, and bootstrap/permutation settings can also change its operating characteristics.

8.3 Small-statistic boundary: noncentral chi-square

A noncentral $\chi^2_\nu(\lambda)$ density is the Poisson mixture

$$f_{\chi^2}(y; \nu, \lambda) = e^{-\lambda/2} \sum_{j=0}^{\infty} \frac{(\lambda/2)^j}{j!} f_{\chi^2}(y; \nu + 2j, 0). \tag{21}$$

For each fixed $\lambda \ge 0$, the terms with $j \ge 1$ vanish relative to the $j = 0$ term as $y \downarrow 0$, and hence

$$\lim_{y \downarrow 0} \frac{f_{\chi^2}(y; \nu, \lambda)}{f_{\chi^2}(y; \nu, 0)} = e^{-\lambda/2}. \tag{22}$$

For a nested central/noncentral pair, this suggests evaluating $m_1(y)/m_0(y)$ through the density *ratio* or its log-domain representation near zero rather than dividing separately evaluated underflowed densities. Implementations should validate quadrature on the nonnegative effect axis, prior normalization, and the limit in Eq. (22). The uploaded source descriptions do not prescribe universal numerical tolerances, permutation seeds, or quadrature grids; those belong in a versioned implementation manifest.

9 Admissibility diagnostics

9.1 Discrete gate

For conditioning-stratum collection $\mathcal{G}(S)$, define $\widehat{H}(X \mid S) = \sum_{g \in \mathcal{G}(S)} (n_g/n) H(X_g)$ and analogously for Y. The discrete gate requires $H(X), H(Y) > \varepsilon_H$ and, when $S \neq \varnothing$, $\widehat{H}(X \mid S), \widehat{H}(Y \mid S) > \varepsilon_H$. It further requires at least $N_{\min} = 1$ stratum containing $m_{\min} = 2$ observations with variation in both endpoints. The documented implementation uses $\varepsilon_H = 10^{-12}$; the optional minimum testable-sample ratio is zero.

9.2 Continuous gate

For n_q finite observations and $d = |S|$, the continuous gate checks

$$n_q \geq \max\{50, 10(d+2)\}, \quad \gamma_S = \frac{\lambda_{\min}(\widehat{\Sigma}_S)}{\lambda_{\max}(\widehat{\Sigma}_S) + \varepsilon} \geq 10^{-6}, \tag{23}$$

with $\gamma_S = 1$ for $S = \varnothing$, and the normalized median nearest-neighbor conditional-variance ratios

$$\rho_X(q) = \frac{\operatorname{median}_i \operatorname{Var}(\{X_j : j \in \mathcal{N}_{20}(i; S)\})}{\widehat{\operatorname{Var}}(X) + \varepsilon} \geq 10^{-3}, \qquad \rho_Y(q) \geq 10^{-3}. \tag{24}$$

For an empty conditioning set, local ratios are defined as one subject to global variation checks. In the final continuous Bayes-PC runs the geometry gate was enabled; every logged tested query passed it. The gate therefore was *not* validated as an independent reliability improvement by these runs. In either data type, inadmissible evidence is recorded but cannot trigger edge deletion or collider orientation.

10 Strong-evidence reliability after adaptive test selection

The finite-sample assessment is separate from the analytical model identities. On 16 discrete benchmark instances, 522 truth-labeled CI queries at conditioning orders 0, 1, and 2 were evaluated by every candidate source; ground truth comes from DAG d-separation. Of these, 234 queries satisfy H_0 and 288 satisfy H_1. At the operational threshold $\tau = 2.3$, define

$$\widehat{\epsilon}_I = \frac{\#\{H_1, \ell > \tau\}}{\#\{H_1\}}, \qquad \widehat{\epsilon}_D = \frac{\#\{H_0, \ell < -\tau\}}{\#\{H_0\}}, \tag{25}$$

with strong coverage the fraction of admissible records outside $[-\tau, \tau]$. These are conditional error frequencies for the sampled truth strata, *not* posterior false-discovery probabilities or universal guarantees.

We also grouped the same 522 query records by the *actually selected* CI test, rather than reassigning them to the best candidate in hindsight. This is a descriptive post-selection analysis, not a randomized comparison between fixed tests. The single false strong-D selected decision is associated with UCI, and the other selected methods show none in this audit. Counts in some selected-method strata are small; they must not be treated as population reliability rankings.

Table 2: Adaptive-selector post-selection errors by conditioning order at $\tau = 2.3$. The final column gives strong-evidence coverage.

Order	H_0	H_1	False strong-I	False strong-D	Coverage
$\|S\| = 0$	56	96	32/96 (33.3%)	1/56 (1.8%)	94.7%
$\|S\| = 1$	82	96	36/96 (37.5%)	0/82 (0%)	90.4%
$\|S\| = 2$	96	96	30/96 (31.3%)	0/96 (0%)	87.5%
Total	234	288	98/288 (34.0%)	1/234 (0.43%)	90.6%

Table 3: Adaptive decisions grouped by the selected discrete CI procedure, pooling the three conditioning orders ($\tau = 2.3$).

Selected test	Queries	H_1	False strong-I	Strong coverage
G^2	117	55	29/55 (52.7%)	88.0%
PLLCIT	64	32	18/32 (56.3%)	95.3%
Pearson χ^2	72	28	13/28 (46.4%)	83.3%
UCI	163	100	25/100 (25.0%)	92.6%
wUCI	106	73	13/73 (17.8%)	92.5%

In particular, the selected-method rates are influenced by which tasks the selector assigns to each method. The audit demonstrates substantial false strong-independence risk under weak or heterogeneous dependence, even though all records pass the admissibility check. A common numeric cutoff alone cannot remove this limitation. Continuous test-specific post-selection reliability is not established by this discrete audit.

For the adaptive source, raising τ from 2.3 to 5.0 reduces the sampled false-strong-I rate from 34.0% to 4.9% while reducing strong-evidence coverage from 90.6% to 42.3%. A corresponding population-conditional guarantee would require additional uniform controls for the fixed queries and the selection mechanism; the empirical rates above do not supply such a theorem.

11 Additional experimental and query-count details

11.1 Threshold sensitivity (final unscreened discrete suite)

Across the 48 unscreened discrete instances, the main threshold comparison is The separate query-

Table 4: Aggregate Bayes-PC sensitivity on the unscreened discrete benchmark (48 instances; per-instance means).

τ	Skeleton F1	Collider F1	CPDAG SHD
1.1	0.885	0.503	5.02
2.3	0.888	0.506	5.06
3.4	0.873	0.528	5.79

level reliability study uses different samples; its threshold-dependent error rates must not be equated with the full-graph recovery metrics above.

11.2 Logical orientation requests

Let C be the fixed learned skeleton, and let $\mathcal{U}$ be its unshielded triples. A CPC-style reference examines endpoint-neighborhood subsets for each endpoint pair, with at most $2^{d_C(X)} + 2^{d_C(Y)}$ sets before de-duplication. In contrast, Bayes-PC processes only triples in $\mathcal{U}^-$ whose stored first-deletion separators exclude the middle node and requests a missing completion $T \cup \{Z\}$ once per stored

base context. For logical orientation requests,

$$Q_{\text{T-PBER}}^{\text{orient}} \leq \sum_{(X,Z,Y)\in\mathcal{U}^-} \left|\{T \in \mathcal{F}_{XY}^-(Z) : T \cup \{Z\} \notin \mathcal{Q}_{XY}^{\text{skel}}\}\right|. \tag{26}$$

Cached logical requests must be distinguished from newly executed CI tests. The frozen-skeleton private-cache experiment in the parent paper reports physical executions and orientation-stage wall-clock separately from read-only cache replay.

12 Reproducibility inventory

To reproduce each $\ell(q)$ exactly, a release needs more than the family of densities given above. It should supply:

- the six frozen selector artifacts, their feature maps, scaler versions, candidate order, and CI task identifiers;
- the definition, normalization and implementation version of each selected test statistic;
- per-test empirical null-fitting and moment-matching code, permutation/bootstrap sampling parameters and random seeds;
- plug-in $\mathcal{C}_{q,k}$ values, positive-effect constraints, numerical quadrature rules, underflow/tail limits and test cases;
- the experimental run commands, software versions, cache policies, hashes of learned skeletons, and raw metrics.

The underlying graduate-study descriptions establish the analytic maps presented in Sections 3–7 but do not, by themselves, verify every implementation-specific clipping tolerance and fitted numerical parameter in the frozen code. Those implementation details should be checked before a permanent public repository release.

Relation to the parent paper. This independent note is intended for repository publication and technical inspection. The conference camera-ready paper contains the principal statistic-level interface, representative Fisher-Z mapping, Bayes-PC method definitions and empirical reliability audit. This file is distributed separately and does not form part of the publisher's 12-page camera-ready PDF.